已知四阶矩阵 A 的特征值分别为 $\pi, -\pi, 0, 0$，求 $\sin A, \cos A$.

1. 解：

设 $\psi(\lambda) = \det(\lambda I - A) = \lambda^2(\lambda-\pi)(\lambda+\pi)$

则根据 Hamilton-Cayley 定理，$\psi(A)=0$

利用待定系数法，设 $\sin\lambda = P_1(\lambda)\psi(\lambda) + a\lambda^3 + b\lambda^2 + c\lambda + d$

则 $\begin{cases} \sin(\pi) = a\pi^3 + b\pi^2 + c\pi + d = 0 \\ \sin(-\pi) = -a\pi^3 + b\pi^2 - c\pi + d = 0 \\ \sin(0) = d = 0 \\ \cos(0) = 3a\cdot 0^2 + 2b\cdot 0 + c = 1 \end{cases}$ 得 $\begin{cases} a = -\frac{1}{\pi^2} \\ b = 0 \\ c = 1 \\ d = 0 \end{cases}$

故由 $\sin\lambda = P_1(\lambda)\psi(\lambda) - \frac{1}{\pi^2}\lambda^3 + \lambda$

$\Downarrow$

$\sin A = -\frac{1}{\pi^2}A^3 + A$

同理，设 $\cos\lambda = P_2(\lambda)\psi(\lambda) + a'\lambda^3 + b'\lambda^2 + c'\lambda + d'$

则 $\begin{cases} \cos(-\pi) = -a'\pi^3 + b'\pi^2 - c'\pi + d' = -1 \\ \cos(\pi) = a'\pi^3 + b'\pi^2 + c'\pi + d' = -1 \\ \cos(0) = d' = 1 \\ -\sin 0 = 3a'\cdot 0^2 + 2b'\cdot 0 + c' = 0 \end{cases}$ 得 $\begin{cases} a' = 0 \\ b' = -\frac{2}{\pi^2} \\ c' = 0 \\ d' = 1 \end{cases}$

故由 $\cos\lambda = P_2(\lambda)\psi(\lambda) - \frac{2}{\pi^2}\lambda^2 + 1$

$\Downarrow$

$\cos A = -\frac{2}{\pi^2}A^2 + 1$

已知 $A=\begin{pmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}$, $f(z)=4+z+6z^3$，求 $f(A)$.

2、解

$\lambda_1=2$, $J_1=\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}$; $\lambda_2=3$, $J_2=\begin{pmatrix} 3 & 1 & \\ & 3 & 1 \\ & & 3 \end{pmatrix}$ 则 $A=\begin{bmatrix} J_1 & \\ & J_2 \end{bmatrix}$

又 $f(J_1)=\begin{bmatrix} f(2) & f'(2) \\ & f(2) \end{bmatrix}=\begin{bmatrix} 54 & 73 \\ & 54 \end{bmatrix}$

$f(J_2)=\begin{bmatrix} f(3) & f'(3) & \frac{f''(3)}{2} \\ & f(3) & f'(3) \\ & & f(3) \end{bmatrix}=\begin{bmatrix} 169 & 163 & 54 \\ & 169 & 163 \\ & & 169 \end{bmatrix}$

$\therefore f(A)=\mathrm{diag}[f(J_1), f(J_2)]=\begin{pmatrix} 54 & 73 & & & \\ & 54 & & & \\ & & 169 & 163 & 54 \\ & & & 169 & 163 \\ & & & & 169 \end{pmatrix}$

已知 $\sin tA = \frac{1}{4}\begin{pmatrix} 4\sin 2t - 12t\cos 2t & 9t\cos 2t \\ -16t\cos 2t & 4\sin 2t + 12t\cos 2t \end{pmatrix}$，试求出矩阵 A 以及 A 的 Jordan 标准型。

3. 解：

$$\frac{d\sin tA}{dt} = \frac{1}{4}\begin{bmatrix} 24t\sin 2t - 4\cos 2t & 9\cos 2t - 18t\sin 2t \\ 32t\sin 2t - 16\cos 2t & 20\cos 2t - 24t\sin 2t \end{bmatrix} = A\cos tA$$

令 $t=0$，则 $A = \frac{1}{4}\begin{bmatrix} -4 & 9 \\ -16 & 20 \end{bmatrix}$

$\det(\lambda I - A) = \begin{vmatrix} \lambda+1 & -\frac{9}{4} \\ 4 & \lambda-5 \end{vmatrix} = (\lambda-2)^2$　　由 $(\lambda I - A) = \begin{bmatrix} 3 & -\frac{9}{4} \\ 4 & -3 \end{bmatrix}$，$r(\lambda I - A) = 1$，故 $\lambda=2$ 代数重度为1

$\therefore$ A的 Jordan 标准型为 $\begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}$

已知 $\boldsymbol{A}\in\mathbf{R}^{m\times n}$，$\boldsymbol{b}\in\mathbf{R}^{m}$．对于矛盾方程组 $\boldsymbol{Ax}=\boldsymbol{b}$，使得 $f(\boldsymbol{x})=\|\boldsymbol{Ax}-\boldsymbol{b}\|_2^2$ 为最小的向量 $\boldsymbol{x}^{(0)}$ 称为最小二乘解．导出最小二乘解所满足的方程组．

4. 解：

$$f(x)=\|Ax-b\|_2^2=(Ax-b,Ax-b)=(Ax-b)^T(Ax-b)$$

$$=(x^TA^T-b^T)(Ax-b)$$

$$=x^TA^TAx-b^TAx-x^TA^Tb+b^Tb$$

$$=x^T(A^TA)x-2(A^Tb)^Tx+b^Tb$$

从而 $\frac{df}{dx}=2A^TAx-2A^Tb=2(A^TAx-A^Tb)$，求极值需令 $\frac{df}{dx}=0$，即 $A^TAx^{(0)}=A^Tb$

又∵ $\frac{d^2f}{dx^2}=2A^TA$．A^TA 为半正定矩阵，故 $x^{(0)}$ 称为 $f(x)$ 的极小值点

即为 $f(x)$ 的最小值点。需满足的方程组为 $A^TAx^{(0)}=A^Tb$

设 $A=\begin{bmatrix}\cos u & -\sin u\\ \sin u & \cos u\end{bmatrix}$，求

(1) $\frac{d}{du}(e^u A)$，$\frac{d}{du}(A^{-1}(u))$；

(2) 若 $u=e^{2t}$，求 $\frac{d}{dt}A(u)$.

5. 解：

$$A=\begin{bmatrix}\cos u & -\sin u\\ \sin u & \cos u\end{bmatrix} \qquad A^{-1}=\begin{bmatrix}\cos u & \sin u\\ -\sin u & \cos u\end{bmatrix}$$

(1) $\frac{d}{du}(e^u A)=\frac{d(e^u)}{du}\cdot A+e^u\cdot\frac{dA}{du}$

$$=\begin{bmatrix}e^u\cos u & -e^u\sin u\\ e^u\sin u & e^u\cos u\end{bmatrix}+\begin{bmatrix}-e^u\sin u & -e^u\cos u\\ e^u\cos u & -e^u\cos u\end{bmatrix}=e^u\begin{bmatrix}\cos u-\sin u & -\cos u-\sin u\\ \cos u+\sin u & \cos u-\sin u\end{bmatrix}$$

$\frac{d}{du}(A^{-1}(u))=-A^{-1}(u)A'(u)A^{-1}(u)$

$$=-\begin{bmatrix}\cos u & \sin u\\ -\sin u & \cos u\end{bmatrix}\begin{bmatrix}-\sin u & -\cos u\\ \cos u & -\sin u\end{bmatrix}\begin{bmatrix}\cos u & \sin u\\ -\sin u & \cos u\end{bmatrix}=\begin{bmatrix}-\sin u & \cos u\\ -\cos u & -\sin u\end{bmatrix}$$

(2) $\frac{d}{dt}(A(u))=\frac{d}{du}(A(u))\cdot\frac{du}{dt}$

$$=\begin{bmatrix}-\sin u & -\cos u\\ \cos u & -\sin u\end{bmatrix}\cdot 2e^{2t}=\begin{bmatrix}-2e^{2t}\sin e^{2t} & -2e^{2t}\cos e^{2t}\\ 2e^{2t}\cos e^{2t} & -2e^{2t}\sin e^{2t}\end{bmatrix}$$

设 $A=\begin{pmatrix}2 & 2 & 1\\ 1 & 3 & 1\\ 1 & 1 & 3\end{pmatrix}$，试用有限待定级数法计算 $\cos A$

6. 解：

$$\det(\lambda I-A)=\begin{vmatrix}\lambda-2 & -2 & -1\\ -1 & \lambda-3 & -1\\ -1 & -1 & \lambda-3\end{vmatrix}=(\lambda-1)(\lambda-2)(\lambda-5)$$

由 Hamilton-Cayley 定理得 $\psi(A)=0$

设 $\cos\lambda=P(\lambda)\psi(\lambda)+a\lambda^2+b\lambda+c$

$$\begin{cases}\cos 1=a+b+c\\ \cos 2=4a+2b+c\\ \cos 5=25a+5b+c\end{cases}\Rightarrow\begin{cases}a=\frac{1}{4}\cos 1-\frac{1}{3}\cos 2+\frac{1}{12}\cos 5\\ b=-\frac{7}{4}\cos 1+2\cos 2-\frac{1}{4}\cos 5\\ c=\frac{5}{2}\cos 1-\frac{5}{3}\cos 2+\frac{1}{6}\cos 5\end{cases}$$

故 $\cos A=aA^2+bA+cI$

$$=\left(\frac{1}{4}\cos 1-\frac{1}{3}\cos 2+\frac{1}{12}\cos 5\right)\begin{pmatrix}7 & 11 & 7\\ 6 & 12 & 7\\ 6 & 8 & 11\end{pmatrix}+\left(-\frac{7}{4}\cos 1+2\cos 2-\frac{1}{4}\cos 5\right)\begin{pmatrix}2 & 2 & 1\\ 1 & 3 & 1\\ 1 & 1 & 3\end{pmatrix}+$$

$$\left(\frac{5}{2}\cos 1-\frac{5}{3}\cos 2+\frac{1}{6}\cos 5\right)\begin{pmatrix}1 & & \\ & 1 & \\ & & 1\end{pmatrix}$$

$$=\begin{pmatrix}\frac{3}{4}\cos 1+\frac{1}{4}\cos 5 & -\frac{3}{4}\cos 1+\frac{1}{3}\cos 2+\frac{5}{12}\cos 5 & -\frac{1}{3}\cos 2+\frac{1}{3}\cos 5\\ -\frac{1}{4}\cos 1+\frac{1}{4}\cos 5 & \frac{1}{4}\cos 1+\frac{1}{3}\cos 2+\frac{5}{12}\cos 5 & -\frac{1}{3}\cos 2+\frac{1}{3}\cos 5\\ -\frac{1}{4}\cos 1+\frac{1}{4}\cos 5 & \frac{1}{4}\cos 1-\frac{2}{3}\cos 2+\frac{5}{12}\cos 5 & \frac{2}{3}\cos 2+\frac{1}{3}\cos 5\end{pmatrix}$$

设 $A=\begin{pmatrix}0.8 & 0\\ 0.4 & 0.5\end{pmatrix}$，证明 $\sum_{k=0}^{\infty}A^k$ 必收敛，并求 $\sum_{k=0}^{\infty}A^k$．

7. 证：

$$\det(\lambda I-A)=\begin{vmatrix}\lambda-0.8 & 0\\ -0.4 & \lambda-0.5\end{vmatrix}=(\lambda-0.8)(\lambda-0.5)$$

$\therefore \rho(A)=\max\{\lambda_i\}=0.8<1 \quad \therefore \sum_{k=0}^{\infty}A^k$ 必收敛

$$\sum_{k=0}^{\infty}A^k=(I-A)^{-1}=\begin{bmatrix}0.2 & 0\\ -0.4 & 0.5\end{bmatrix}^{-1}=\begin{bmatrix}5 & 0\\ 4 & 2\end{bmatrix}$$

证明 $f(A^T)=[f(A)]^T$，并利用此结果计算 $\sin tA, e^{tA}$，其中

$$A=\begin{pmatrix} 1 & & & \\ 1 & 1 & & \\ & 1 & 1 & \\ & & 1 & 1 \end{pmatrix}。$$

8. 证：

设A以Jordan分解写成 $A=TJT^{-1}$

则 $f(A^T)=f((TJT^{-1})^T)=f((T^{-1})^T J^T T^T)=(T^{-1})^T f(J^T) T^T$

∵ $J=\mathrm{diag}(J_1, J_2 \cdots J_n)$ $\quad f(J)=\mathrm{diag}(f(J_1), f(J_2) \cdots f(J_n))$

又∵ $f(J_i^T)=\begin{pmatrix} f(\lambda_i) & & & \\ f'(\lambda_i) & f(\lambda_i) & & \\ \vdots & & \ddots & \\ \frac{f(\lambda_i)}{(n-1)!} & \cdots & & f(\lambda_i) \end{pmatrix}=[f(J_i)]^T$ $\quad$ ∴ $f(J^T)=[f(J)]^T$

∴ $f(A^T)=(T^{-1})^T f(J^T) T^T=(T^{-1})^T f(J)^T T^T=(Tf(J)T^{-1})^T=[f(A)]^T$

$\Rightarrow f(A^T)=[f(A)]^T$

故 $\sin tA=(\sin tA^T)^T=\begin{pmatrix} \sin t & t\cos t & -\frac{t^2}{2}\sin t & -\frac{t^3}{6}\cos t \\ & \sin t & t\cos t & -\frac{t^2}{2}\sin t \\ & & \sin t & t\cos t \\ & & & \sin t \end{pmatrix}^T=\begin{pmatrix} \sin t & & & \\ t\cos t & \sin t & & \\ -\frac{t^2}{2}\sin t & t\cos t & \sin t & \\ -\frac{t^3}{6}\cos t & -\frac{t^2}{2}\sin t & t\cos t & \sin t \end{pmatrix}$

$e^{At}=(e^{A^Tt})^T=\begin{pmatrix} e^t & te^t & \frac{t^2}{2}e^t & \frac{t^3}{6}e^t \\ & e^t & te^t & \frac{t^2}{2}e^t \\ & & e^t & te^t \\ & & & e^t \end{pmatrix}^T=\begin{pmatrix} e^t & & & \\ te^t & e^t & & \\ \frac{t^2}{2}e^t & te^t & e^t & \\ \frac{t^3}{6}e^t & \frac{t^2}{2}e^t & te^t & e^t \end{pmatrix}$